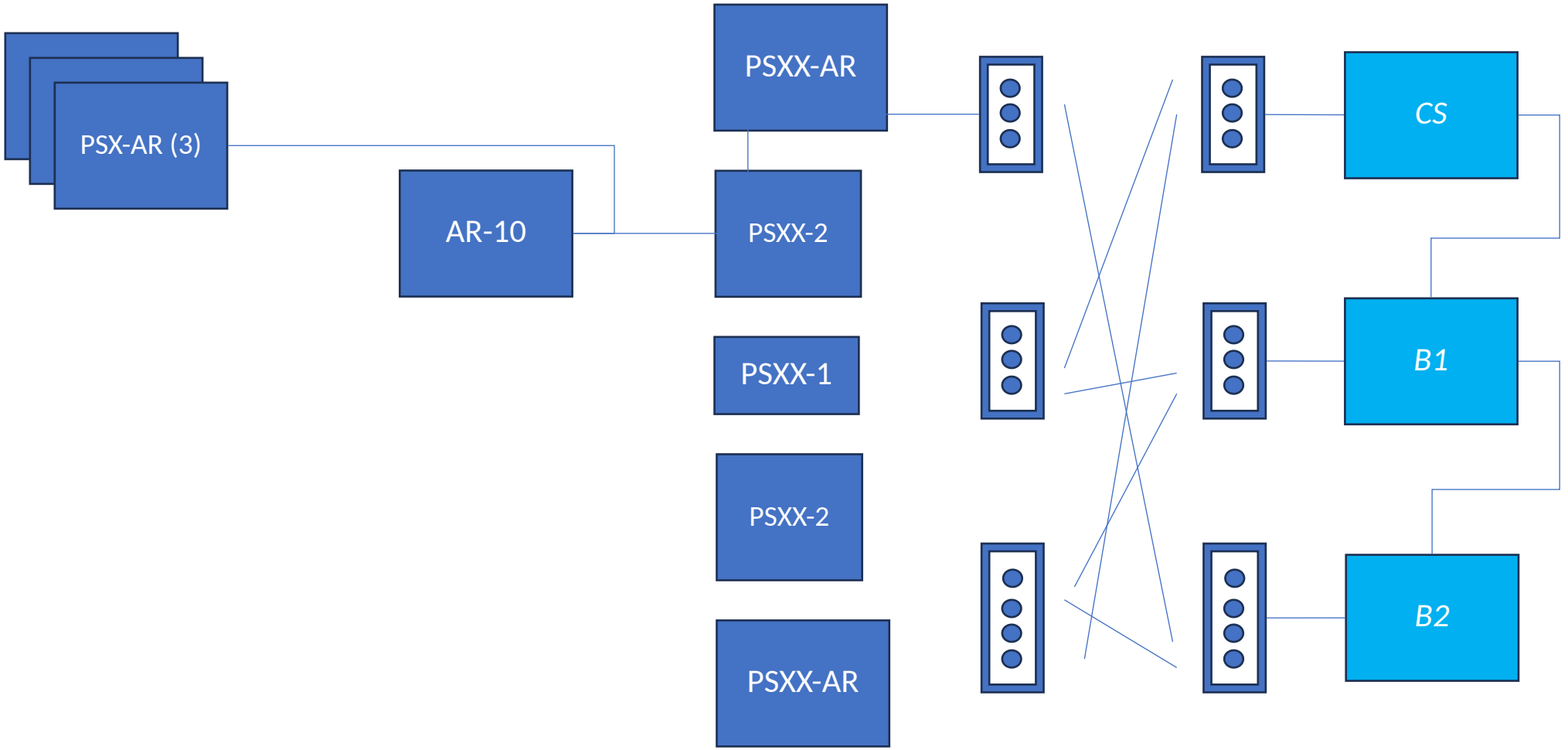
Track Busses							
#	Booster-Bus		Bus Colors				
	Location	Name	North	South	Location	Hardware	Type
1	Helix - Upper	NIL	Red	Black	Neville's Island Loop	PSXX-AR	Auto-Reverser
2	Helix - Upper	BL	Red	Black	Branch Line	PSXX-2	Circuit-Breaker
3	Helix - Upper	ML	White	Blue	Main Line	PSXX-2	Circuit-Breaker
4	Helix - Upper	PCC	White	Blue	Pittsburgh Coke & Chemical	PSXX-1	Circuit-Breaker
5	Helix	HX	Red	Black	Helix	PSXX-2	Circuit-Breaker
6	Helix	SCY	White	Blue	Scully Yard	PSXX-2	Circuit-Breaker
7	Helix	ERZ	Red	Black	East Loop	PSXX-AR	Auto-Reverser
8	West	SRZ	Red	Black	Scully Reverse Zone	AR-10	Auto-Reverser
9	Peninsula - Upper	BRZ	Red	Black	Branch Loop	PSXX-AR	Auto-Reverser
10	Peninsula - Upper	CIZ	White	Blue	Canonsburg Inner	PSXX-AR	Auto-Reverser
11	Peninsula - Upper	COZ	White	Blue	Canonsburg Outer	PSXX-AR	Auto-Reverser

Power Districts



Circuit Breaker ← → Power Booster

Matrix

Terminal Bus Description

DCC1 – Red/Black for primary bus.
Green/Brown for secondary

Bus Wiring

PS Output

DCC – Control Bus

DNCT2B

PSXX-2
PSXX-1

CS

B1

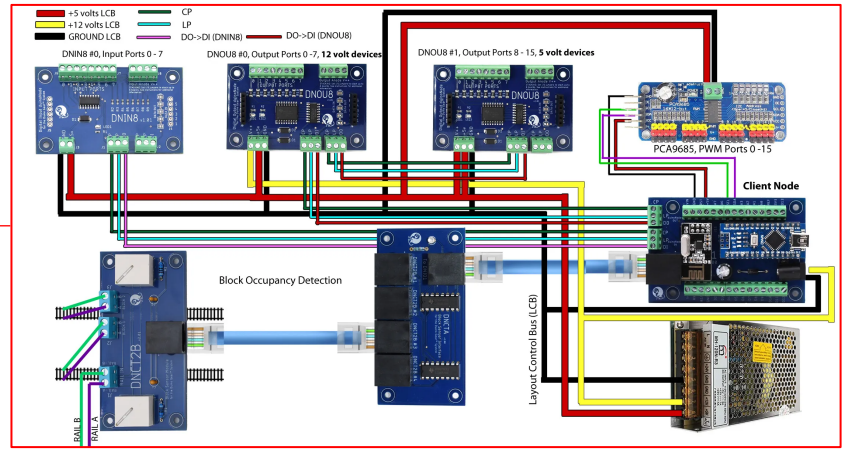
B2

DNCTA

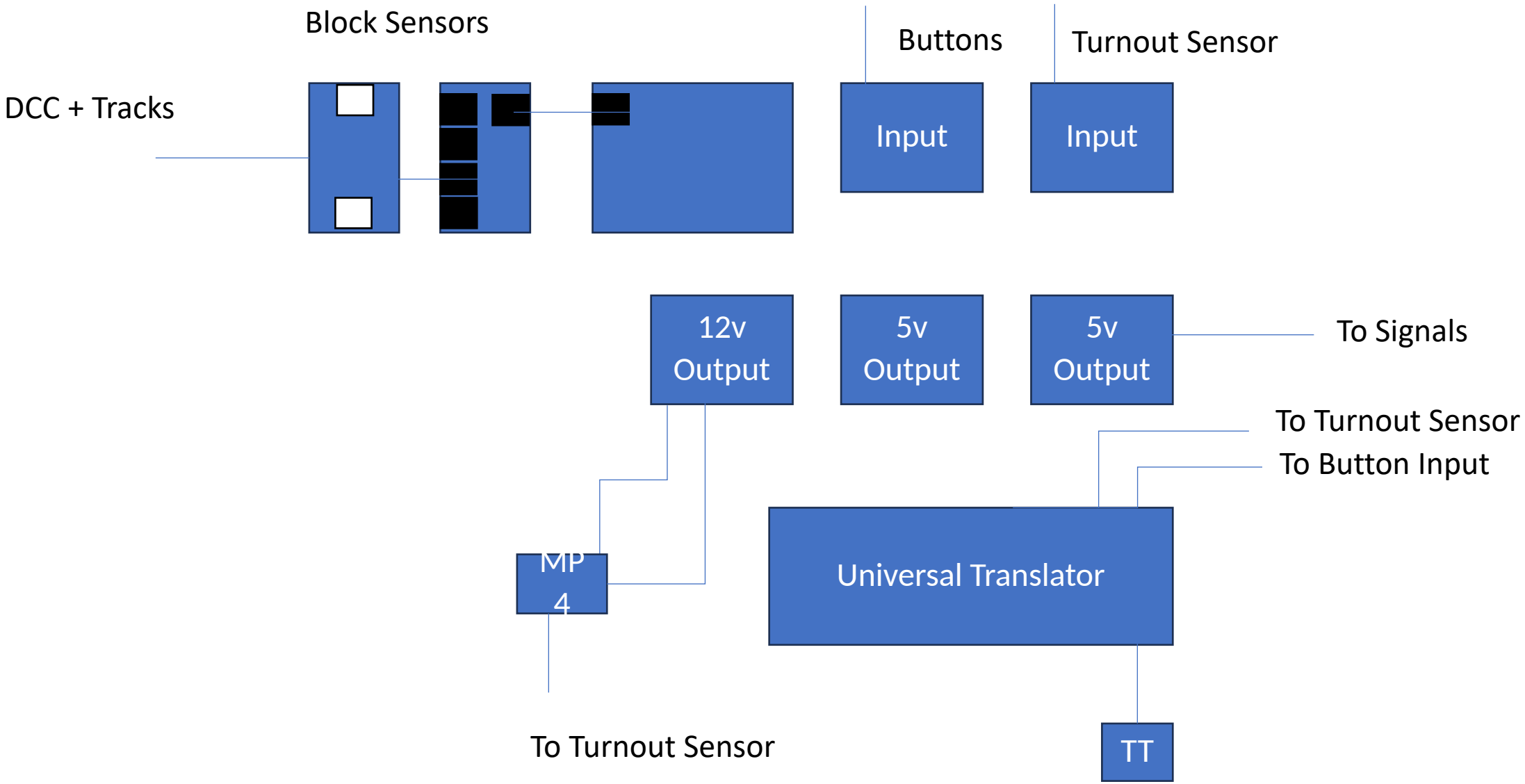
DNCT2B

LCN

+12V LCB



Control Node Configuration

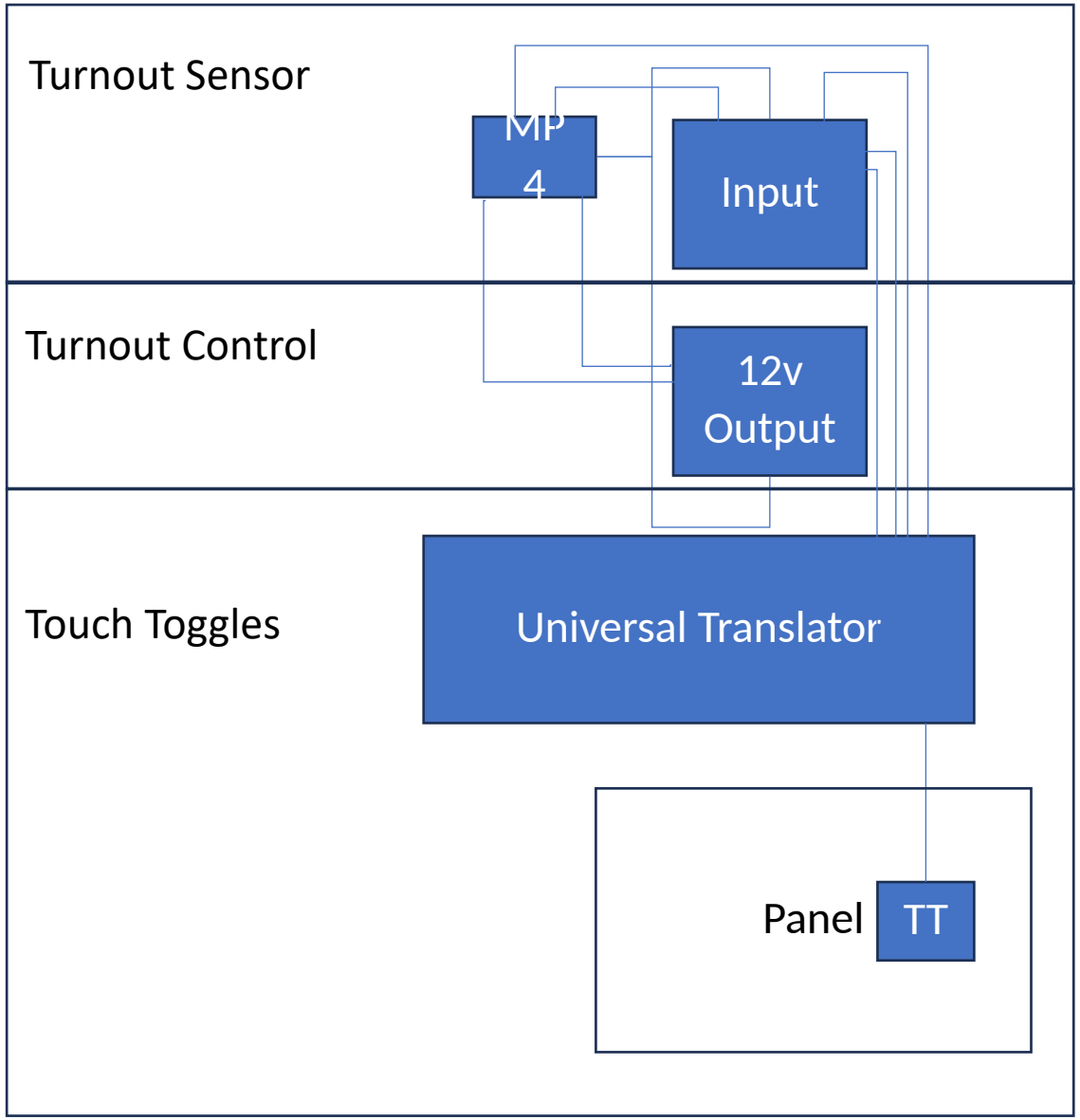


The Aux relay on the MP4 switches is used to connect to the input ports on the DNIN8 as feedback for Normal and Reverse positions. The NC position should be used to drive the input indication on the universal translator.

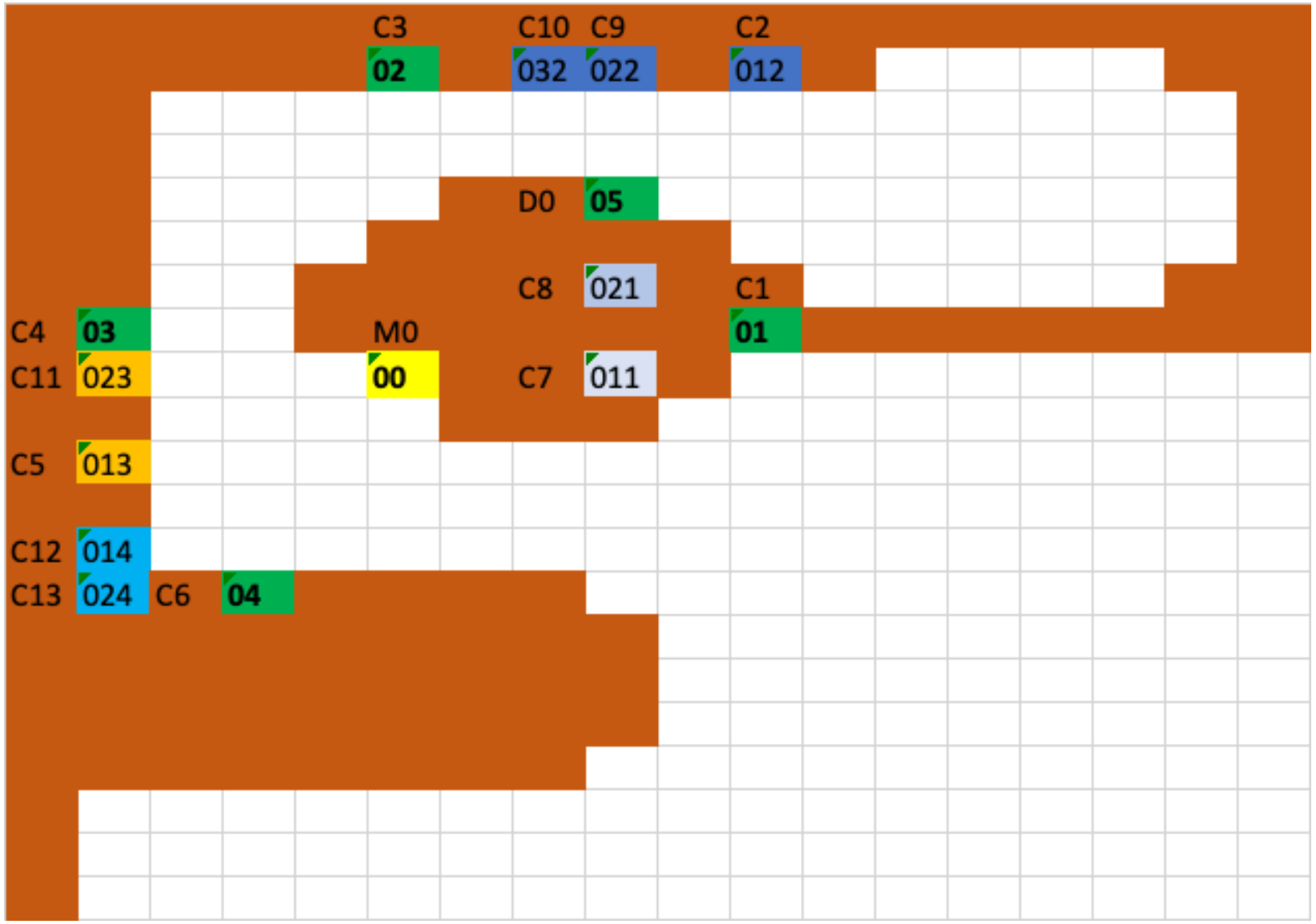
A separate pair of inputs connect the switch contacts to an input on the DNIN8 to detect when the turnout button is pressed.

The universal translator allows the indication on the touch toggle to be represented by the position of the motor. This is necessary when also having DCC control of the turnout.

Turnout Wiring



Control Node Placement



Protocols

Digital Command Control (DCC)
DCC sends both power and digital commands through the rails, giving each locomotive independent control. A DCC bridge passes accessory commands to LCN so turnouts and automation can be operated from both throttles and JMRI.

JMRI (Java Model Railroad Interface)
JMRI is the software hub for decoder programming, dispatching, and control panels. It controls turnouts via DCC and receives LCN state through MQTT for real-time block, turnout, and signal feedback.

LCN (Layout Control Nodes)
LCN is a wireless, distributed control system running LCOS. Nodes placed near devices handle block detection, turnouts, signals, lighting, and automation. The DCC bridge ensures accessory commands are shared between JMRI and handheld throttles.

MQTT
MQTT is the messaging layer linking JMRI and LCN. A custom LCOS node will broadcast LCN state, allowing JMRI to capture turnout commands and sensor feedback.